

Micturition Disorders



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KEYWORDS

• Incontinence • Dysuria • Urethral obstruction • Overactive bladder • Urine retention

KEY POINTS

- Differentiation of conscious versus unconscious voiding is important to determine the cause of micturition disorders.
- It is important to determine whether the clinical signs are caused by a disorder of storage or a disorder of voiding.
- Observation of the patient while urinating is key to diagnosing disorders of voiding.
- Patients with disorders of voiding have difficulty fully emptying the bladder and can have overflow incontinence.
- Patients with disorders of storage can have urinary incontinence or increased frequency of urination caused by decreased storage capability.



Video of functional outflow obstruction accompanies this article at www.vetsmall.theclinics.com/

INTRODUCTION

Physiology of Micturition

The primary purpose of the lower urinary tract is to store urine and to facilitate its elimination at an appropriate time. The bladder is in storage phase 99% of the time and in emptying phase only 1% of the time. Coordination of storage and emptying requires a complex interaction between the somatic and autonomic nervous systems, as well as normal function of the organs and tissues involved.

All 3 components of the peripheral nervous system are involved in the micturition cycle (**Table 1**). In addition, conscious voiding involves the lumbar and sacral spinal cord as well as the brainstem and cerebral cortex (**Fig. 1**).

During filling and storage, stretch receptors in the bladder wall send afferent signals along the pelvic nerve, which activate a reflex arc through the hypogastric nerve to the urethra (**Fig. 2**). Norepinephrine is released by postganglionic neurons to activate

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Table 1
Peripheral nervous system components of micturition

Type (Receptor)	Location	Nerve	Function When Stimulated	Function When Blocked	Function When Inappropriately Stimulated	Function When Inappropriately Blocked
Parasympathetic (M ₃ muscarinic)	Bladder body (detrusor)	Pelvic nerve (S1–S3)	Contraction and bladder emptying	Detrusor relaxation and bladder filling	Overactive bladder	Bladder atony, urine retention
Sympathetic (beta ₃ -adrenergic)	Bladder body (detrusor)	Hypogastric nerve (L1–L4)	Detrusor relaxation and filling	Detrusor contraction and urination	Urine retention	Decreased bladder compliance and increased filling pressure
Sympathetic (alpha ₁ -adrenergic)	Bladder neck/urethra	Hypogastric nerve (L1–L4)	Contraction and continence	Urination	Urine retention	Open urethra, incontinence
Somatic (nicotinic)	Distal urethra/pelvic floor	Pudendal nerve (S1–S2)	Conscious/reflex contraction and continence	Urination	Urine retention	Open urethra, incontinence

Storage Phase

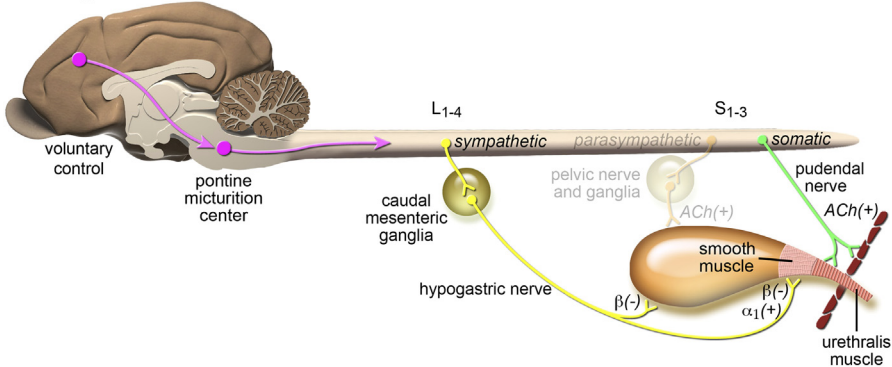


Fig. 1. Innervation and signal pathways during the storage phase of the micturition cycle. The symbol (+) denotes stimulation of muscular contraction, and (–) denotes inhibition of muscular contraction. ACh, acetylcholine-mediated receptors; α , alpha-adrenergic receptors; β , beta-adrenergic receptors; L_{1–4}, lumbar spinal cord segments 1 to 4; S_{1–3}, sacral spinal cord segments 1 to 3. (Drawing by Tim Vojt, MA. Used with permission from The Ohio State University.)

beta₃-adrenergic receptors in the bladder wall, allowing relaxation and continued filling. Norepinephrine also stimulates alpha₁-adrenergic receptors in the urethra and causes contraction of the urethral smooth muscle, thus preventing urine leakage. During initiation of micturition, stretch receptors send afferent signals along myelinated fibers of the pelvic nerve to the lumbar spinal cord and cranial to the pontine micturition center in the brain. Signals from the cerebral cortex and the hypothalamus are processed to determine whether the situation is appropriate for urination. If so, signals are sent down the pelvic nerve, which leads to acetylcholine release at the postganglionic parasympathetic neurons. Acetylcholine binds to M₃ receptors and stimulates bladder smooth muscle contraction. At the same time, inhibitory signals are sent to the sympathetic reflexes and the urethra relaxes, allowing normal emptying.

A local urethral reflex arc is the somatic-mediated contraction of the striated muscle surrounding the urethra in response to a sudden increase in abdominal pressure and movement of urine into the urethra, such as during a cough or sneeze.

Voiding Phase

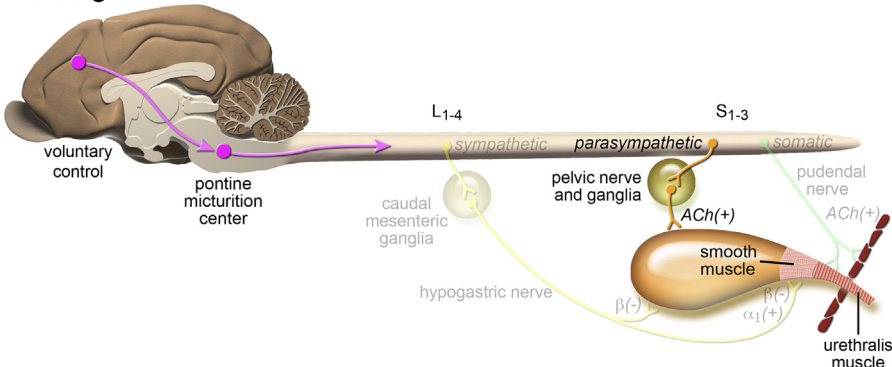


Fig. 2. Innervation and signal pathways during the voiding phase of the micturition cycle. (Drawing by Tim Vojt, MA. Used with permission from The Ohio State University.)

In addition to normal neurologic mechanisms, several other factors are important for the normal function of the micturition cycle in dogs and cats. The integrity of the smooth muscle of the urethra, normal urethral mucosa that creates a watertight seal, associated vasculature, and the support of connective tissues are also key.^{1,2} In females, estrogen seems to have a significant impact on these tissues.^{3,4} Its decline after neutering seems to play a role in the urethral sphincter mechanism incompetence (USMI) seen in neutered female dogs.^{5,6} Alternatively, prostatic hyperplasia in intact male dogs can lead to functional urethral obstruction and urinary retention.

Disorders of Storage

Storage disorders primarily occur because of an inability to maintain adequate urethral tone if there are normal or increased bladder pressures. These disorders may result from an anatomic or developmental abnormality or acquired dysfunction in the spinal cord, urinary bladder, or urethra and surrounding tissues. Disorders of storage generally result in the unconscious loss of urine: urinary incontinence. These disorders include USMI, lower motor neuron bladder (LMB), and overactive bladder (OAB).

Urethral Sphincter Mechanism Incompetence

USMI is the most common storage disorder in dogs.⁷ The urethral sphincter mechanism involves the smooth muscle of the urethra as well as the surrounding support tissues, submucosal vasculature, and urothelium. USMI is thought to result from the breakdown in this complex through reduced muscular responsiveness and tone, and structural changes in the periurethral tissues. In female dogs these changes seem to be associated with a reduction in estrogen and alterations in follicle-stimulating hormone and luteinizing hormone after neutering; hence the term hormone-responsive incontinence.^{8,9} The pathophysiology of its development in male dogs is poorly understood; however, there may be a relationship with decreased testosterone levels, and increased follicle-stimulating hormone and luteinizing hormone levels, because it appears more frequently in neutered than in intact males.^{10,11}

USMI is most common among neutered female dogs. It is less common in neutered males and rare in cats.¹¹ It seems to affect up to 20% of neutered females and up to 30% of large breed dogs weighing more than 20 kg. Several breeds have been found to have increased risk, including the Doberman pinscher, giant schnauzer, Old English sheepdog, rottweiler, Weimaraner, and boxer, but any breed may be affected.¹² Recent literature has shown an increased risk of development of USMI in dogs neutered before 3 months of age.^{13,14} However, this finding remains controversial because other studies have found no relationship with age at neuter.¹⁵ It is possible that this is related to the development and sensitivity of the sphincter mechanism tissues to estrogen and other hormones, but the mechanism is incompletely understood.

Predisposing factors such as a pelvic bladder, short urethra, or recessed vulva may also be associated with increased risk of developing USMI; however, the presence of one of these does not necessarily lead to incontinence. A reduction in pressure transmission from the abdomen to the proximal urethra has the potential to lead to urinary incontinence in animals with a short urethra or pelvic bladder by setting up a negative pressure gradient from the bladder to the urethra.

Clinical Presentation and Diagnosis of Urethral Sphincter Mechanism Incompetence

Signs of incontinence may begin at any time, but appear most often within a few years of neutering. Dogs with USMI usually have normal bladder capacity and are able to urinate normally with complete emptying of the bladder. Although most of these dogs are otherwise healthy, clinical signs can worsen dramatically if comorbidities

develop, leading to polyuria or difficulty posturing to urinate. The increased volumes of urine in the bladder, and for longer periods of time, may lead to increased pressure on the already weakened sphincter mechanism and leakage. Lower urinary tract infection (UTI) can also lead to increased frequency of clinical signs. It is suspected that USMI, and urinary incontinence of any cause, is associated with increased risk of UTI; however, no studies have been conducted to fully evaluate this.

Owners often seek veterinary care when the frequency becomes bothersome, although, in the author's experience, the incontinence may have been present for months or years. It is essential to establish that the animal is passing urine unconsciously. Many owners complain of incontinence when the animal is showing submissive urination or is otherwise consciously urinating in inappropriate places. Dogs with USMI may leak urine when recumbent, sleeping, or after exertion. Although the passage of urine is involuntary, dogs may increase grooming of the perivulvar or preputial area. USMI is an acquired condition, so questioning of the owner regarding the onset of clinical signs is important. Animals that have been incontinent from birth or before neutering should be assessed for congenital malformations such as ectopic ureters before making a diagnosis of USMI.

Physical examination of these animals is often normal, with the exception of perivulvar or preputial urine staining. As mentioned earlier, the conformation of the vulva should be noted, although its contribution to USMI is unclear. The presence of perivulvar dermatitis, regardless of conformation, may increase the risk of UTI. Rectal examination and palpation of the urethra are usually normal. Urination is observed to be normal and complete, with little or no residual volume. The assessment of residual urine is essential in male dogs to differentiate USMI from detrusor urethral dyssynergia, which is a disorder of incomplete bladder emptying (discussed later).

The presence of urinary incontinence in an otherwise healthy neutered female dog that was previously continent is often adequate for presumptive diagnosis of USMI and a trial of empiric therapy. Because of its lower incidence, intact females, intact or neutered males, and cats with similar histories and clinical signs require additional evaluation before making a diagnosis of USMI.

The animal should be observed urinating, if possible. Normal posturing and a full stream without stranguria should be noted. Assessment of residual urine volume may be made by bladder catheterization or estimated using abdominal ultrasonography of the bladder. A urinalysis and urine culture should be performed. The presence of isosthenuria or hyposthenuria may contribute to severity of incontinence and may require additional evaluation for an underlying cause. Animals with polyuria may not respond to medical treatment as well as those making normal volumes of urine. As noted earlier, the risk of UTI in animals with USMI has not been evaluated, but its presence may lead to worsening signs. Therefore, screening for, and treating of, an infection may increase therapeutic success. A complete blood count and serum biochemistry panel may be indicated but are not always necessary for diagnosis of USMI, although they may be helpful when making therapeutic choices, and are important in evaluating animals with polyuria.

If anatomic abnormalities are suspected, imaging, such as contrast radiography, abdominal ultrasonography, or contrast computed tomography, is indicated. Cystoscopic evaluation of the lower urinary tract allows assessment of the entire urethra, ureter placement, and bladder mucosa. It also allows detailed investigation of vestibular and vaginal conformation. Advanced diagnostics, such as urodynamic studies, are designed to quantify the pressure produced along the urethra and the compliance and detrusor function of the urinary bladder. These diagnostics are available at many referral institutions and are most often indicated in patients with equivocal signs and poor response to therapy.

Medical Treatment of Urethral Sphincter Mechanism Incompetence

Medical therapy for USMI is generally considered the first line of management; only after failure or intolerance of medical therapy are surgical options considered. Medical treatment of USMI in neutered female dogs consists of increasing the number and sensitivity to estrogen of alpha receptors in the urethral sphincter, or stimulating those receptors with an alpha-agonist (Table 2).

The most commonly used estrogens are diethylstilbestrol (DES) and estriol (Incurin, Merck Animal Health, Madison, NJ). DES is not available commercially so it must be compounded. Adverse effects associated with estrogen use include mammary gland development, vulvar swelling, and attractiveness to males.¹⁶ These effects are usually dose related and subside with dose reduction. A more serious adverse event associated with estrogen use is irreversible bone marrow suppression.¹⁷ It is considered standard of care to monitor the complete blood count in animals receiving estrogen compounds; however, the doses most associated with bone marrow suppression are much higher than those recommended to treat USMI. The author recommends performing a complete blood count before starting treatment with an estrogen compound, and then rechecking it 1 month into treatment. Estriol seems to have a higher response rate among female dogs than DES (89% and 65% respectively).^{16,18}

Phenylpropanolamine (PPA; Proin, PRN Pharmacal, Pensacola, FL) is the most widely used alpha-agonist for the treatment of USMI. It is commercially available in sizes designed for use in dogs. The dose and frequency needed for each animal tend to vary widely and may need to be increased over time to maintain continence. Adverse effects associated with PPA include restlessness, aggression, changes in sleeping patterns, and gastrointestinal signs. These effects are also usually alleviated by a reduction in dose or frequency.^{19,20} Clinical response to PPA administration ranges from 75% to 90%.^{20,21} Both an estrogen and PPA are often used in the same patient for severe or refractory incontinence. There is little evidence supporting a synergistic increase in efficacy; however, there are anecdotal reports of greater improvement than on a single-medication regimen.²²

PPA is also frequently used in male dogs and cats with USMI. In addition, neutered male dogs may be treated with testosterone cypionate as monthly injections however its efficacy is not well documented. Although male dogs with USMI are most responsive to PPA, it is at a rate of only 43% showing good to excellent response, which is much lower than in female dogs.¹¹ Estrogen compounds should not be used in males because of the risk of prostatic metaplasia and feminization. They are rarely used in cats, and care should be taken to monitor the mammary glands for neoplasia in cats receiving estrogens.

Surgical Treatment of Urethral Sphincter Mechanism Incompetence

In patients that fail medical therapy, it may be necessary to consider surgical intervention. Surgery is generally only pursued if the animal does not respond to, or cannot tolerate, medical treatment. Several surgical procedures have been used to treat USMI, many based on the theory of increasing the transmission of abdominal pressure to the proximal urethra. These procedures include colposuspension, transobturator vaginal tape, and urethropexy. These procedures have had variable outcomes and are considered to have poor long-term efficacy, particularly in animals with normal bladder position.²³ The most promising surgical procedure for USMI is placement of an artificial urethral sphincter that may be adjusted via a subcutaneous port. Recent studies have shown it to lead to a significant increase in continence in male and female dogs that had failed medical therapy for USMI.^{24–26}

Table 2
Drugs frequently used to treat USMI

Drug	Class	Dose	Side Effects	Caution
Phenylpropanolamine	Alpha-agonist	1.0–1.5 mg/kg PO q 8–12 h	Hypertension, aggression, restlessness, gastrointestinal upset, anxiety	Hypertension, hyperadrenocorticism, renal disease
DES	Estrogen	0.1–1.0 mg/dog PO q 24 h for 5–7 d then weekly or as needed	Myelosuppression (rare at these doses), attractiveness to males, mammary/vulvar swelling, behavior changes	Males (can develop prostatic metaplasia)
Conjugated estrogen	Estrogen	0.02 mg/kg PO q 24 h for 5–7 d then q 2–4 d as needed	Same as DES	Same as DES
Estriol	Estrogen	2 mg/dog PO q 24 h × 14 d, then reduce to 1 mg q 24 h	Same as DES	Same as DES
Testosterone cypionate	Androgen	2.2 mg/kg IM q 4–8 wk	Behavior change, aggression, perianal adenoma, prostatic hyperplasia	Cardiac disease, renal disease, hepatic disease, prostatic disease

Abbreviations: DES, diethylstilbestrol; IM, intramuscular; PO, orally; q, every.

Data from Papich MG. Saunders handbook of veterinary drugs, small and large animal. 3rd edition. Philadelphia: Elsevier; 2010.

In the past, injectable urethral bulking agents have been used, particularly bovine cross-linked collagen, to increase resting urethral pressure in dogs with USMI. In 2009 the manufacturers of the most commonly used product removed it from the market, so, at the time of writing, injectable bulking agent therapy for USMI is considered unavailable, except in some research studies.

Lower Motor Neuron Bladder

Incontinence as a disorder of storage can also occur secondary to spinal cord injury or disease. Lesions in the S1 to S2 region lead to weakness of the striated muscular sphincter. Disruption of the local reflex arc at these segments leads to an easily expressible bladder that may empty with minor increases in abdominal pressure. These animals are identified by decreased anal tone and a poor perineal reflex as well as their easily expressible bladder. Most of these animals are unable to voluntarily void and require intermittent catheterization or manual expression by the owner. Correction of the underlying cause may lead to some return to normal function. Because of the tendency to have incomplete emptying of the bladder with manual expression, these animals are at increased risk of developing UTI and appropriate monitoring must be in place. The muscarinic agonist bethanechol has been used in these patients to increase detrusor contraction; however, the evidence for its efficacy remains controversial.^{27,28}

Detrusor Hyperreflexia/Overactive Bladder

Detrusor hyperreflexia/OAB is the most common form of urinary incontinence in people, but it has been poorly characterized in domestic species. The incidence of OAB and its importance as a cause of urinary incontinence in dogs and cats are unknown. In people it is characterized by sudden urgency to urinate and involuntary loss of urine associated with bursts of detrusor contractions at bladder volumes far less than capacity. In dogs it may manifest as an animal that loses bladder compliance and capacity and thus may need to urinate more often than before without an increase in urine production or inflammation of the lower urinary tract. OAB may be a cause of treatment failure in some dogs treated for USMI, and should be considered in these cases. Diagnosis of OAB in animals can be challenging and is only definitively made using urodynamic studies such as cystometrography. Response to therapy with antimuscarinic drugs is often used to presumptively diagnose OAB in veterinary species. The most commonly used include oxybutynin and imipramine, the latter of which also has alpha-agonist effects, potentially increasing urethral sphincter tone. Adverse effects of these medications include gastrointestinal abnormalities such as diarrhea and constipation, as well as parasympatholytic signs like tachycardia and hyposalivation.

Disorders of Emptying

The inability to completely empty the bladder during a normal void can result from either a functional or mechanical obstruction of the outflow tract and urethra, or an abnormality of the detrusor muscle that impairs its complete contraction. Overflow incontinence can result from the lack of complete emptying, often when the animal is at rest. An owner may not be able to differentiate this from an animal with a storage disorder (eg, USMI), and it is up to the clinician to determine the underlying process so that appropriate therapy can be instituted. Inability to completely empty the bladder is a risk factor for UTI, and these animals should be monitored for infection.

Detrusor Atony

Complete emptying of the urinary bladder depends on the normal contraction of the detrusor muscle. In the normal animal, the bladder relaxes during filling with only small increases in intravesical pressure. As the bladder continues to fill, the pressure increases to a threshold that triggers a detrusor contraction and emptying. Loss of adequate detrusor contraction can result from neurogenic or nonneurogenic abnormalities. Injury to the sacral spinal cord S1 to S3 or pelvic nerves can lead to bladder atony, and is often associated with weakened urethral tone. These animals often have decreased perineal reflexes and easily expressed bladders (LMB). Treatment of the underlying lesion, if possible, may lead to improved voiding function. However, until normal function returns, careful management of urination must be followed. Male dogs are manually expressed or aseptically catheterized 2 to 4 times a day either in the hospital or at home, and cats and female dogs are often manually expressed.

Direct damage to the detrusor muscle can occur from overdistension caused by mechanical or functional outflow obstruction of an acute or chronic nature. The muscle fibers of the detrusor transmit action potentials that initiate contraction via tight junctions. With overdistension, these tight junctions are interrupted, leading to an absent or ineffective contraction. The overdistension may be acute, as with obstructive feline idiopathic cystitis in a male cat, or chronic, as with a dog with a functional obstruction of the urethra. Relief of the obstruction and maintenance of a small bladder volume for up to 2 weeks may allow the junctions to reestablish and the return of coordinated detrusor function. This condition is usually managed by either indwelling or frequent sterile catheterization of the bladder. Bethanechol may be used in these patients to enhance stimulation of the detrusor contraction because the pelvic nerve is intact. Other medications that have been shown to enhance detrusor function are cisapride and metoclopramide; however, the individual response varies widely.^{29,30} It is essential that relief of any urethral obstruction, functional or mechanical, be attained before starting medical therapy to enhance detrusor contraction.

Functional Obstruction: Detrusor Urethral Dyssynergy

Functional urethral obstruction, or detrusor urethral dyssynergy (DUD), arises from an abnormality in the reflex arc that allows the urethral sphincter to relax at the initiation of detrusor contraction and urination. The lesion is thought to be in the reticulospinal tract, Onuf nucleus, or the caudal mesenteric ganglion, and it is possible that the lesion involves the loss of inhibitory signals to the pudendal and hypogastric nerves.³¹ It is unknown whether there is a more local lesion to the nerves, the neuromuscular junction, or the smooth or striated urethral sphincters.³² Unlike the upper motor neuron bladder seen in animals with thoracolumbar intervertebral disc disease and other spinal cord lesions, these animals typically have an otherwise normal neurologic examination.

The disorder affects primarily middle-aged large and giant-breed male dogs, although female dogs and cats can be affected. One case series of 22 dogs reported a mean age of 4.9 years.³³ Clinical signs are similar to those of mechanical obstruction. The animal often postures to urinate and is able to produce a urine stream that quickly becomes attenuated or stops completely. The animal may continue to posture to urinate or make several attempts without fully emptying the bladder ([Video 1](#)). The presence of large amounts of residual urine typically leads to overflow incontinence and may be mistaken for USMI. This leakage can occur because the hypertonicity of the involved sphincter is often dynamic and is triggered by the act of voiding. In chronic cases, bladder overdistension and subsequent atony may develop. Unlike animals with mechanical obstruction, these dogs are typically easy to catheterize and

usually do not show increases in urethral pressure on urodynamic evaluation unless actively voiding. Contrast urethrography may be normal or reveal areas of narrowing of the urethra (urethrosperm).

Presumptive diagnosis of DUD is often made by observing the dog urinate with a typical interrupted pattern, documentation of a large residual urine volume, easy passage of a urinary catheter, and ruling out of a mechanical obstruction. Normal residual urine volumes in 48 normal dogs were reported to be 0.1 to 3.4 mL/kg body weight with a mean of 0.2 mL/kg.³⁴ The author uses less than 0.5 mL/kg as a general guideline. Ultrasonography is recommended to assess the ureters and renal pelvises for dilatation secondary to chronic obstruction and ureterorenal reflux of urine. Additional diagnostics, including contrast urethrography, urethrosperm, or urodynamic evaluation, may be necessary to verify the diagnosis in patients that fail to respond to medical therapy.

Treatment of the hypertonic urethral sphincter generally consists of alpha-adrenergic blockade with prazosin, an alpha₁-specific antagonist with demonstrated effects on both the internal and external urethral sphincters.³⁵ Tamsulosin, which is specific for the alpha_{1A} subtype found in the internal urethral sphincter, has also been successful in these dogs. Some dogs require additional therapy if the striated muscle is more significantly affected. Benzodiazepines, such as diazepam, or other skeletal muscle relaxants, including acepromazine and methocarbamol, may be more effective if the external urethral sphincter is involved. Diazepam is typically administered 30 minutes before voiding to decrease external urethral sphincter pressure. Dantrolene and baclofen have been used in the past as skeletal muscle relaxants; however, the potential for adverse effects has decreased their use in veterinary patients.^{35,36} In severe and refractory cases, intermittent sterile catheterization by the owner at home may be necessary. Medical therapy for associated bladder atony should only be started after adequate relief of the functional urethral obstruction has been reached (Table 3). Close monitoring of these patients for residual urine volume and UTI is needed to assess the efficacy of treatment and prevent complications.

Prognosis for recovery of normal voiding is good, but most dogs require lifelong therapy for DUD. Attempts to taper medications to the lowest effective dose may be hampered by relapse of clinical signs after months of normal voiding.³³ Prognosis seems to be worse in patients with bladder atony or UTI secondary to urine retention. Anecdotally, urethral stenting may improve clinical signs; however, this is considered a salvage procedure with several potential complications, and is indicated only in the most refractory of cases.

Functional Obstruction: Upper Motor Neuron Bladder

Neurogenic functional urethral obstruction is generally caused by spinal cord lesions cranial to the sacral segment, which leads to loss of inhibitory signals to the hypogastric and pudendal nerves, which prevents sphincter relaxation on voiding. This condition is the classic upper motor neuron bladder in which the patient is unable to urinate normally and is difficult to manually express. The most commonly affected patients are those with intervertebral disc disease and associated paresis. These animals typically have additional neurologic deficits, including paresis and nociceptive loss. Treatment of the underlying lesion typically leads to partial or complete return to normal voiding function after days to weeks. Until normal voiding resumes, the patients are managed as for DUD with alpha-adrenergic blockade and manual expression or catheterization. Monitoring for overdistension and UTI is critical in these patients, as is nursing care, particularly because the overflow incontinence that can accompany this process may lead to skin breakdown in these recumbent patients.

Table 3
Drugs frequently used to treat disorders of emptying

Drug	Mechanism	Dose	Side Effects/Caution
Prazosin	Alpha ₁ -antagonism, smooth muscle relaxant	1 mg/animal <15 kg q 8–12 h PO 2 mg/animal >15 kg q 8–12 h PO	Hypotension, weakness, syncope, GI upset/renal disease, cardiac disease
Tamsulosin ^a	Alpha _{1A} -antagonism, smooth muscle relaxant	0.01–0.2 mg/kg PO q 24 h	Hypotension
Phenoxybenzamine	Nonspecific Alpha-antagonism, smooth muscle relaxant	Dog: 0.25 mg/kg q 8–12 h Cat: 1.25–7.5 mg/cat q 8–12 h PO	Hypotension, tachycardia, miosis/first dose hypotension
Acepromazine	Nonspecific Alpha-antagonism Skeletal muscle relaxant Anxiolytic	Dog: 0.5–2.2 mg/kg q 6–8 h PO Cat: 1.1–2.2 mg/kg q 6–8 h PO	Sedation, hypotension
Diazepam	Skeletal muscle relaxant Anxiolytic	Dog: 0.5–2 mg/kg q 8 h PO or 30 min before voiding	Sedation, ataxia/liver dysfunction Do not use in cats
Methocarbamol	Skeletal muscle relaxant	22–44 mg/kg q 8 h PO	CNS sedation, weakness, hypersalivation
Baclofen ^a	Skeletal muscle relaxant	Dog: 1–2 mg/kg q 8 h PO	Weakness, GI upset; do not use in cats
Dantrolene	Skeletal muscle relaxant	Dog: 1–5 mg/kg q 8 h PO Cat: 1–2 mg/kg q 8 h PO	Weakness/liver disease
Bethanechol	Parasympathomimetic	Dog: 2.5–15 mg/dog q 8 h PO Cat: 1.25–5 mg/cat q 8 h PO	Diarrhea/GI or urethral obstruction
Cisapride	Prokinetic	Dog: 0.1–0.5 mg/kg q 8–12 h PO Cat: 2.5–5 mg/cat q 8–12 h PO	Ataxia, GI upset/GI obstruction

Abbreviations: CNS, central nervous system; GI, gastrointestinal.

^a From Lane IF, Westropp JL, Urinary incontinence and micturition disorders: pharmacologic management, Kirk's Current Veterinary Therapy. 14th edition. St Louis (MO): Elsevier; 2009. p. 955–9.

Data from Papich MG. Saunders handbook of veterinary drugs, small and large animal. 3rd edition. Philadelphia: Elsevier; 2010.

Dysautonomia

Dysautonomia is a rare condition involving the degeneration of the neurons of the autonomic nervous ganglia, leading to sympathetic and parasympathetic dysfunction. It has been most commonly reported in both dogs and cats in the United Kingdom and Scandinavia, but small clusters of cases in both species have been seen in the United State.³⁷ Its underlying cause is unknown, but neurotoxin exposure is strongly suspected. In addition to the classic findings of unresponsive mydriasis and prolapsed nictitans, the abnormalities seen with dysautonomia can range from ileus and constipation to decreased systolic function of the heart. These dogs and cats often have significant voiding dysfunction and urine retention caused by bladder atony and overflow incontinence secondary to urethral sphincter incompetence. There is no specific treatment of dysautonomia and spontaneous remission is uncommon. The prognosis is related to the severity of the clinical signs and the specific organ functions affected. Management of the urinary complications is usually possible, involving bladder expression and/or urethral catheterization, along with parasympathomimetic agents. It is the challenging management of the gastrointestinal components of the disease that often leads to euthanasia.

SUMMARY

The accurate determination of a micturition disorder as one of storage or voiding is essential to selecting appropriate medical and surgical therapy. Differentiation between conscious and unconscious voiding, as well as observation of the patient urinating, can greatly increase success in diagnosis and management of these disorders. Therapy often relies on pharmacologic manipulation of neurologic control of storage and voiding, with surgical options pursued if medical management is unsuccessful.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found online at <http://dx.doi.org/10.1016/j.cvsm.2015.02.006>.

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